



Veeam Backup Engineer Program

Design, Protect & Secure Enterprise Workloads with Veeam Backups



SMART MATE VENTURES

Enterprise IT Training & Certification



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About Smart Mate Ventures

Our Mission

To deliver world-class enterprise IT training that bridges the gap between theoretical knowledge and real-world infrastructure deployments — empowering professionals to build, protect, and recover critical workloads.

Our Vision

To become the most trusted name in enterprise backup, virtualization, and cloud training across the IT industry.

Our Value

We don't just teach technology—we build confidence, practical skills, problem-solving abilities, and successful IT careers.

Enterprise Learning Approach

Curriculum modeled after industry leaders like VMware, Veeam, and AWS

Real Lab Experience

Hands-on access to live enterprise environments

Industry Trainers

Instructors with 10+ years of production infrastructure experience

Career & Interview Prep

Resume review, mock interviews, and certification guidance



Why Choose This Program

A comprehensive skill path covering every dimension of enterprise backup and data protection engineering.



Enterprise Labs

Simulate real-world production environments



Cloud Fundamentals

S3, Azure Blob, Cloud Connect



Backup & Storage

Full coverage of repos, tiers & jobs



Career Guidance

Industry-aligned career roadmaps



Virtualization

Deep VMware & Hyper-V integration



Networking

Transport modes, WAN optimization



Certification

Exam-ready preparation & guidance



Interview Prep

Mock interviews & L2/L3 Q&A banks

Enterprise Backup & DR Fundamentals



What You'll Learn

- Backup vs Replication · DR vs Business Continuity
- RPO & RTO definitions and real-world application
- Backup types: Full, Incremental, Synthetic Full, Reverse Incremental
- 3-2-1-1-0 Rule · Air-Gapped & Immutable Backup
- Enterprise backup challenges and lifecycle management



Lab: Draw a Backup Architecture · Define RPO/RTO · Create a Backup Strategy for a Small Business

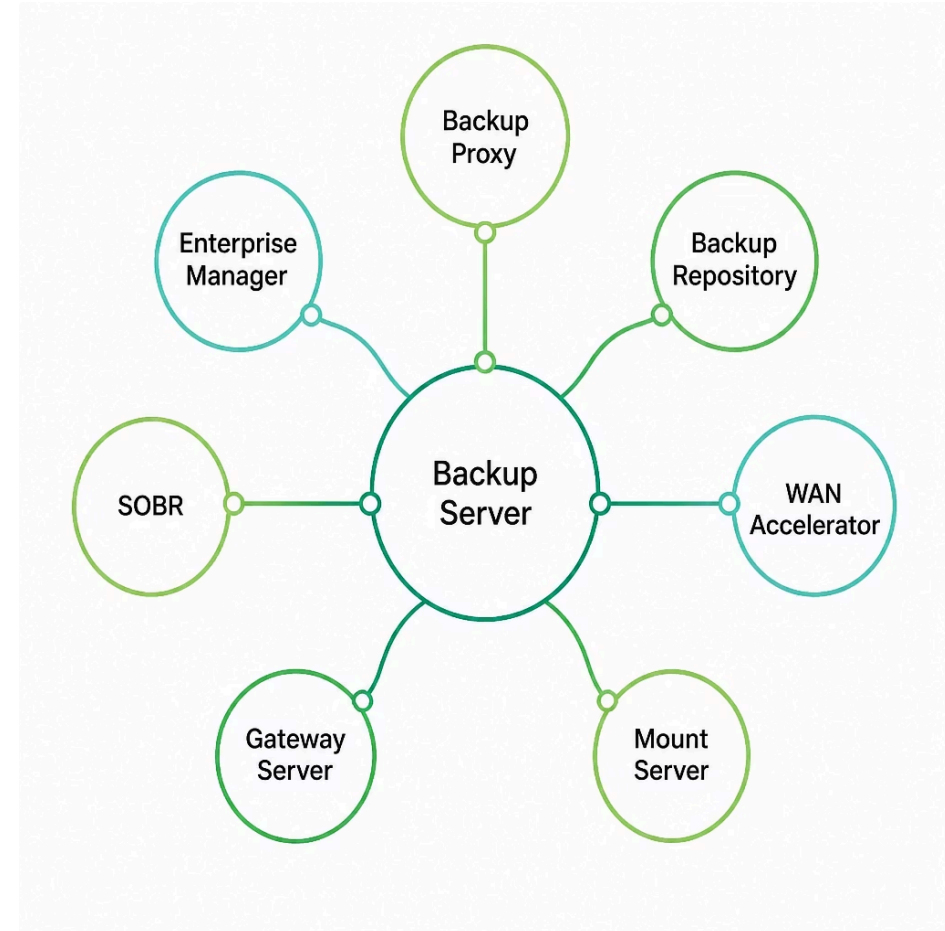
Veeam Backup & Replication v13 Architecture

Core Components

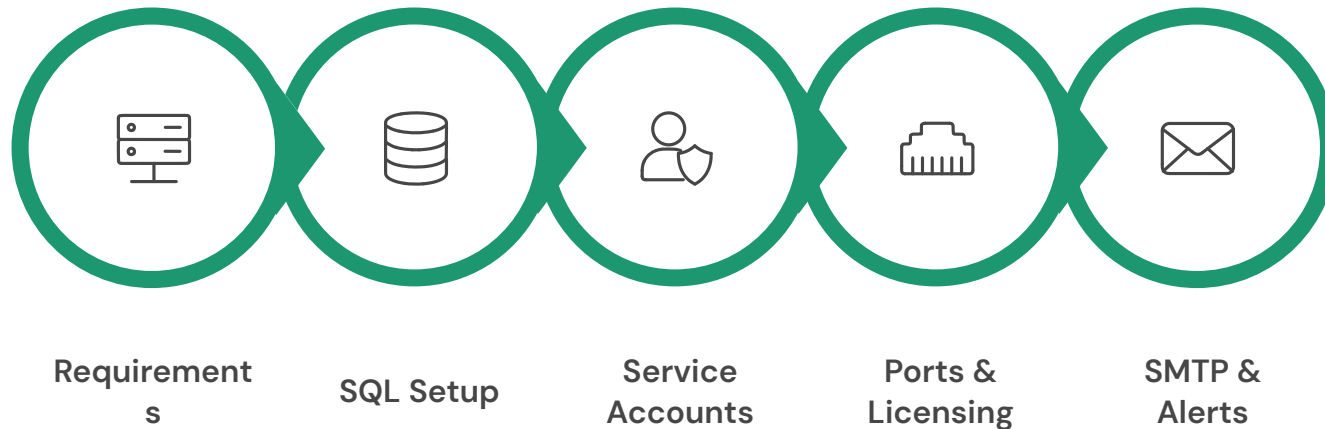
- Backup Server, Proxy, Repository, Mount Server
- WAN Accelerator · Gateway Server
- Enterprise Manager · Backup Console
- Scale-Out Backup Repository (SOBR)
- Veeam Backup Appliance



Lab: Install VBR · Explore Console · Configure License · Understand Component Roles



Installation & Initial Configuration



Key Topics

- SQL Express vs SQL Server — when each is appropriate
- PostgreSQL Database
- Service account design and port communication matrix
- Licensing models and update/patch management

  **Lab: Deploy Veeam Server ·**
Install SQL · Apply License ·
Configure SMTP Notifications



MODULE 4

VMware & Hyper-V Integration

vCenter & ESXi Integration

Add and manage vCenter and ESXi hosts · Configure VM Discovery for automated inventory

VMware CBT

Changed Block Tracking for efficient incremental backups · Snapshot management and reset procedures

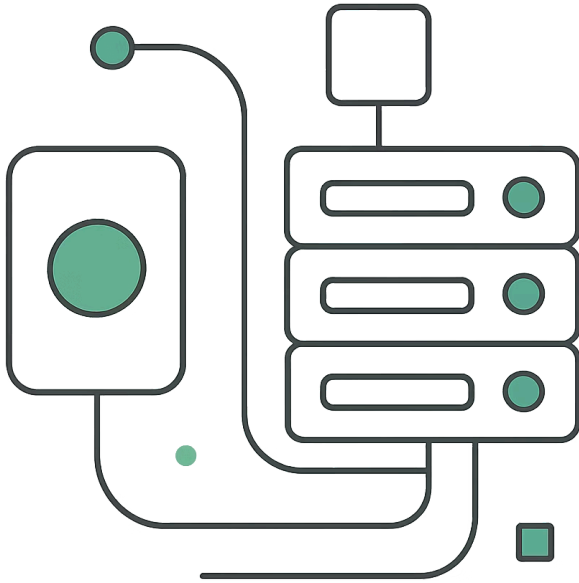
Guest Processing

Application-aware processing for consistent backups of SQL, Exchange, and Active Directory workloads



Lab: Add vCenter · Add ESXi · Verify CBT · Configure Guest Processing

Backup Repository Management



Repository Types & Tiers

- **Windows Repository** — Standard SMB/NTFS-based storage
- **Linux Hardened Repository** — Immutable XFS-based protection
- **CIFS Repository** — Network share integration
- **Deduplication Appliances** — DDR/HPE StoreOnce
- **Performance · Capacity · Archive Tiers** — SOBR tiering strategy



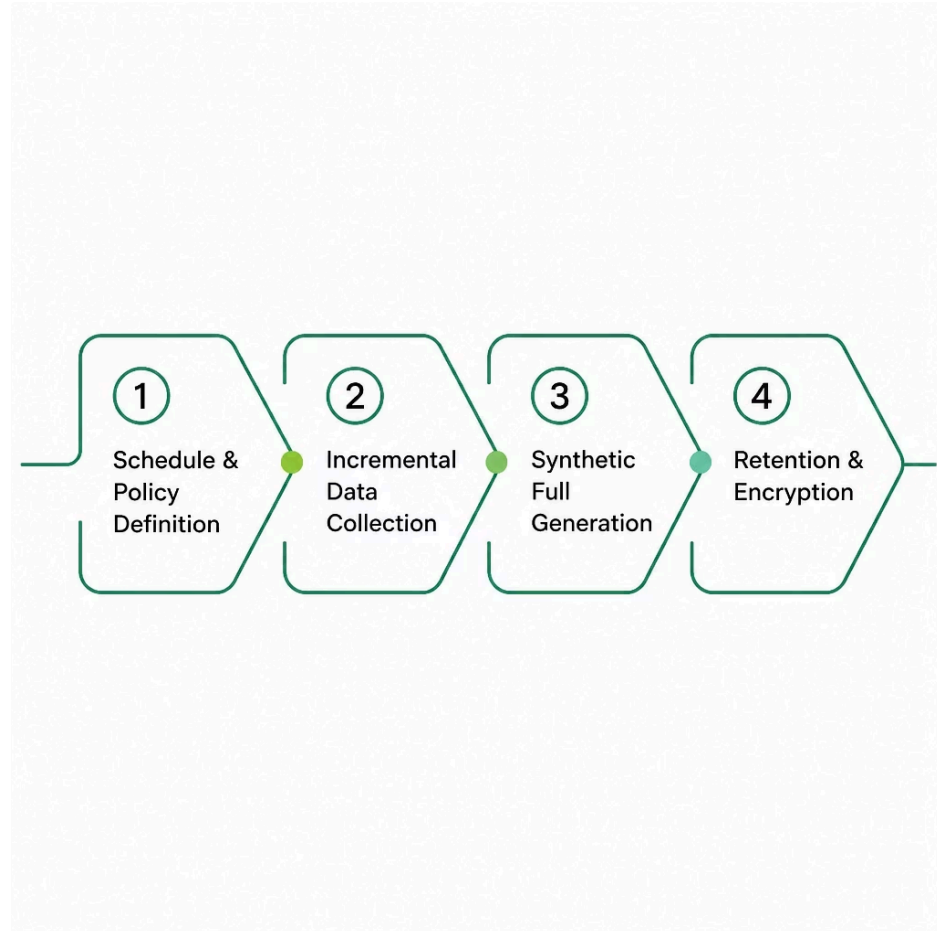
Lab: Configure Windows Repo · Deploy Linux Hardened Repo · Enable Immutability · Set Permissions

Backup Jobs Administration

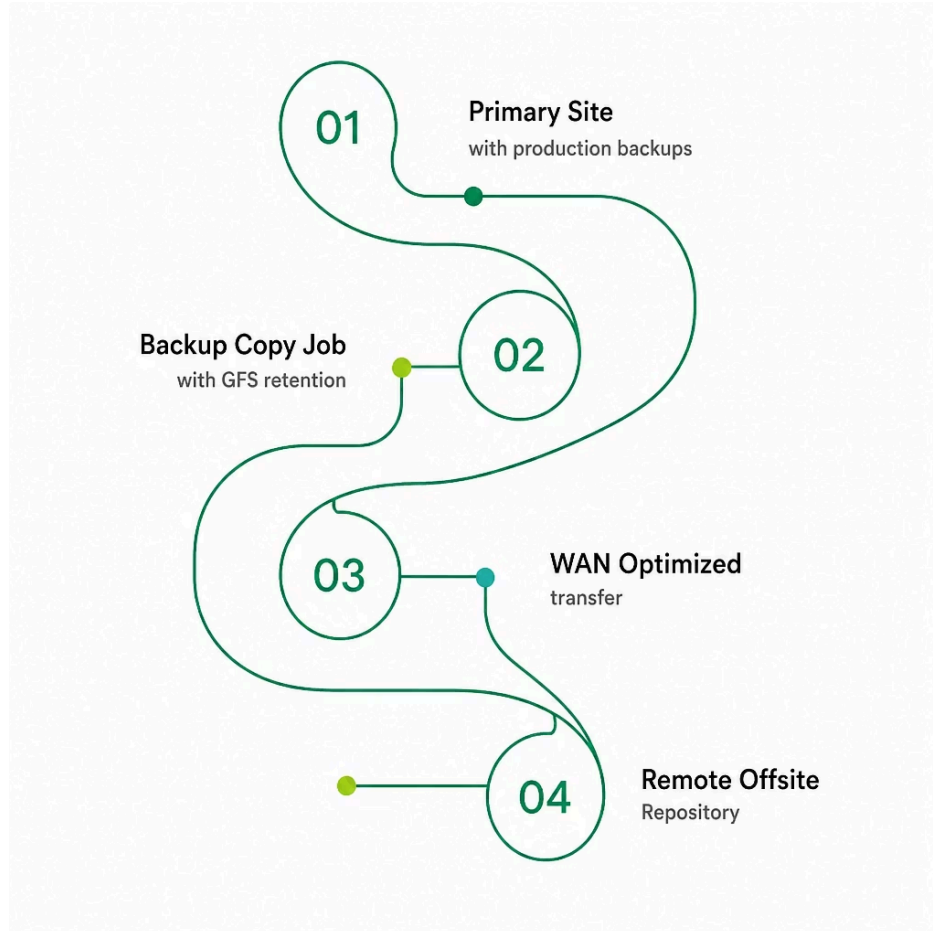
Job Configuration & Policies

- VM Backup Jobs
- Backup Policies
- Scheduling
- Retention Policies
- Incremental Chains
- Synthetic Full vs Active Full backups
- Compression levels
- Backup encryption

 **Lab:** Create Daily Backup Job · Configure Weekly Synthetic Full · Enable Encryption · Monitor Jobs



Backup Copy Jobs

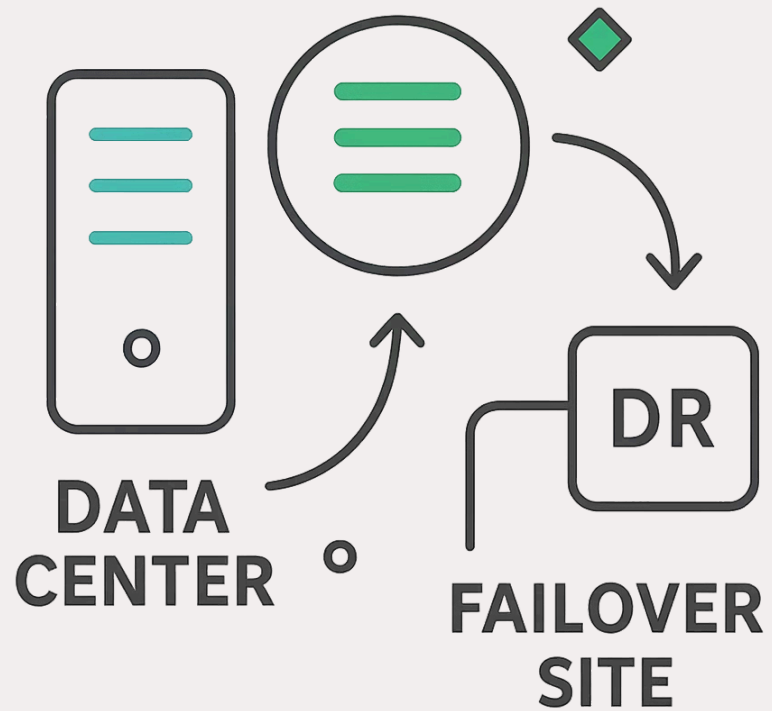


Offsite Data Protection

- Requirement of Backup Copy Job
- Creation of Backup Copy Job
- Secondary copies for GFS retention (daily/weekly/monthly/yearly)
- Copy modes: Immediate vs Periodic
- WAN Optimization for bandwidth-constrained sites
- Offsite backup validation and restore point management



Lab: Create Backup Copy Job · Configure GFS · Verify Restore Points



MODULE 8

Replication

VM Replication

Configure replica jobs to secondary or DR sites with near-zero RPO for mission-critical VMs

Failover Options

Planned Failover · Test Failover
· Permanent Failover — choose the right method for each scenario

Failback & Re-IP

Return workloads to production and reconfigure network settings automatically using Re-IP Rules



Lab: Configure Replication · Test Failover · Execute Failback

Restore Operations

Recovery Methods

→ Instant VM Recovery

Boot VMs directly from backup with minimal RTO

→ File-Level & Guest Restore

Granular recovery of individual files and folders

→ Application Item Recovery

SQL, Exchange, Active Directory object-level restore

→ Secure Restore & Malware Detection

Scan restore points before recovery to prevent reinfection



Lab: Restore Deleted File · Instant VM Recovery · SQL DB Restore · AD Object Restore

SOBR & Object Storage Integration

Scale-Out Backup Repository

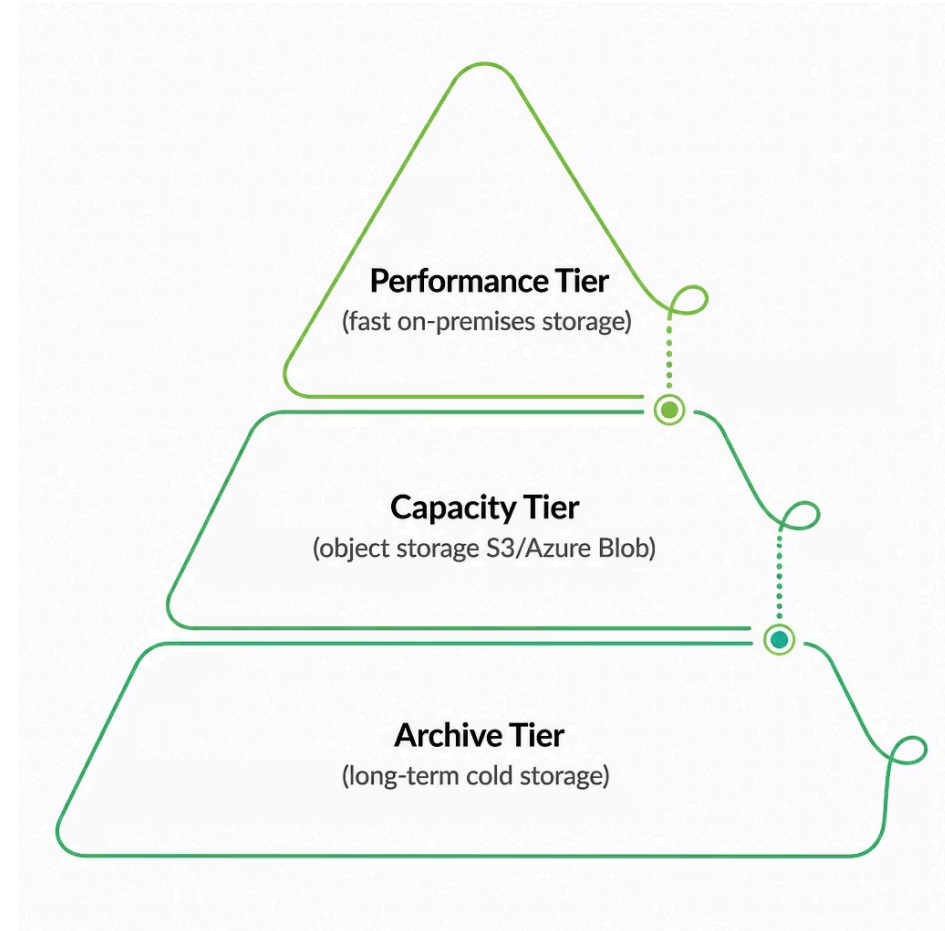
- SOBR Architecture with Performance, Capacity & Archive Tiers
- Copy Mode vs Move Mode for data offloading
- Object storage as Capacity Tier extension

Object Storage Targets

- Amazon S3
- Azure Blob
- Wasabi
- S3-Compatible
- Object Lock & Immutability for ransomware protection



Lab: Configure Object Storage · Enable Immutability ·
Test SOBR Offloading



Linux Hardened Repository



1

Deploy Ubuntu Server

Provision a dedicated Linux host with minimal attack surface and no GUI

2

Configure XFS & Fast Clone

Format storage with XFS filesystem to enable reflink-based Fast Clone technology

3

Enable Immutability

Activate single-use credentials and immutable flag to prevent ransomware from deleting backups

4

Validate Backup Protection

Apply security hardening best practices and verify immutability enforcement

Veeam Enterprise Manager



Centralized Management at Scale

Veeam Enterprise Manager provides a unified web-based console for managing multiple Veeam Backup & Replication servers across distributed environments.

- **Self-Service Portal** — Delegate restore tasks to VM owners
- **Reporting** — Consolidated compliance and SLA dashboards
- **Delegated Administration** — Role-based access control
- **Restore Portal** — End-user file recovery without IT involvement



Lab: Install Enterprise Manager · Add Backup Server · Configure Restore Portal

Security & Ransomware Protection



Zero Trust Architecture

MFA enforcement · Role-based access · Least-privilege design



Air Gap & Immutability

Offline backup copies · Object Lock · Hardened Linux repos



Malware Detection

Threat Hunter scans · Secure Restore · Encrypted backup chains



Lab: Enable Encryption · Configure Secure Restore · Scan Restore Points · Test Malware Detection



Veeam ONE Monitoring



Proactive Visibility Across Your Environment

Veeam ONE delivers real-time infrastructure monitoring, predictive capacity analytics, and automated alerting — ensuring SLA compliance and early issue detection.

- Health checks
- Performance trending
- Capacity forecasting
- Pre-built and custom alert policies
- SLA compliance reporting per job and workload



Lab: Install Veeam ONE · Configure Monitoring · Create Custom Alerts · Generate Reports

SureBackup & SureReplica

Automated Backup Verification

SureBackup eliminates uncertainty by automatically booting VMs in an isolated Virtual Lab and running a suite of tests — all without impacting production.

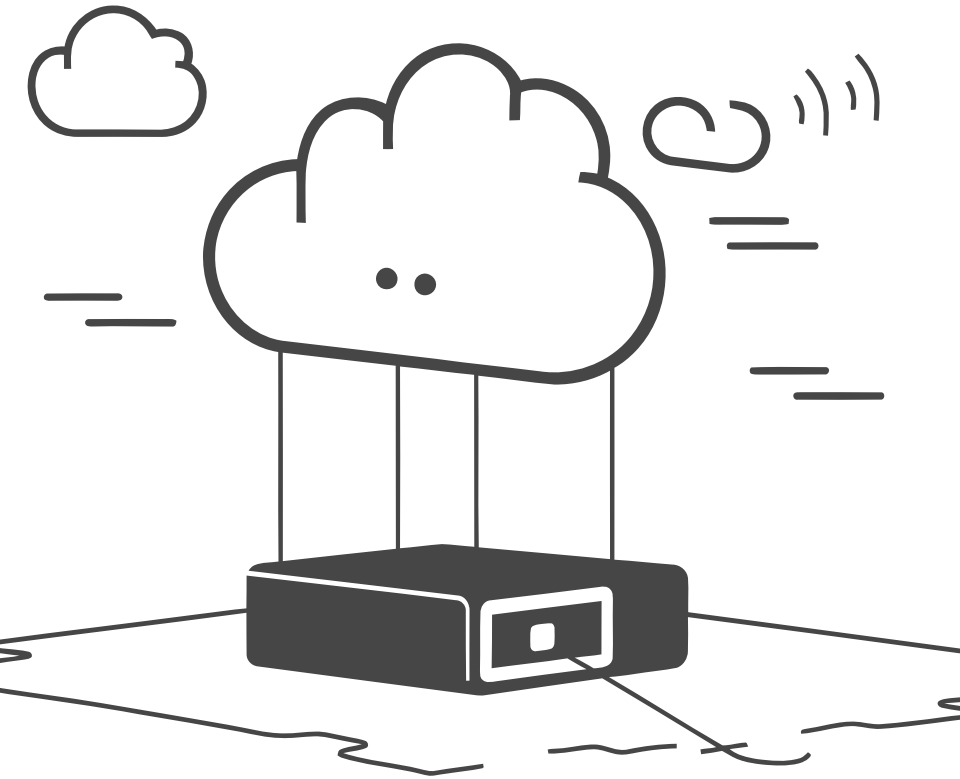
- **Heartbeat Test** — Confirm VM OS boots successfully
- **Ping Test** — Verify network stack responsiveness
- **Application Test** — Script-based service validation
- **SureReplica** — Same validation for replica VMs at DR site



Lab: Configure Virtual Lab · Create SureBackup Job · Validate Recovery



NAS Backup & Cloud Connect



NAS Backup

- SMB & NFS file share backup with versioning
- Flexible restore: full share, folder, or individual file

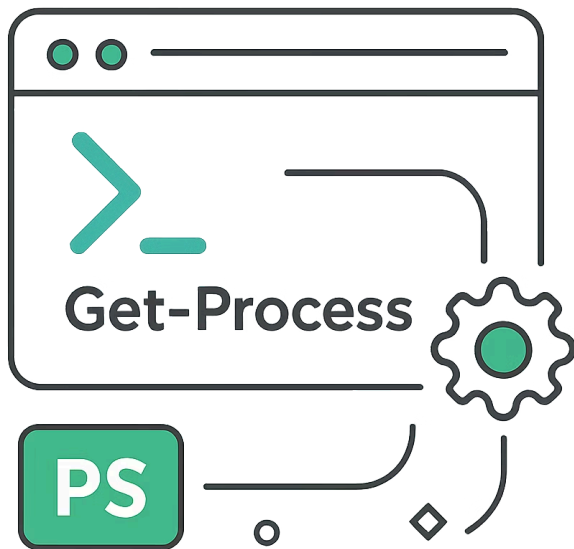
 **Lab:** Add NAS Share · Create Backup Job · Restore Files

Veeam Cloud Connect

- Service Provider architecture for BaaS/DRaaS delivery
- Tenant configuration
- Cloud Repository
- Cloud Replication

 **Lab:** Configure Cloud Connect · Add Tenant · Test Connectivity

Veeam Automation



Automate at Enterprise Scale

Integrate Veeam with your operational workflows using native automation tools — reducing manual effort and enabling infrastructure-as-code practices for backup.

- **PowerShell Module** — Full job lifecycle management via CLI
- **REST API** — Integrate with ITSM, Ansible, or custom portals
- **Scheduled Reports** — Automated compliance and status exports



Lab: Connect via PowerShell · Generate Backup Report ·
Start/Stop Jobs · Export Inventory



MODULE 20

Performance Tuning

Proxy & Repository Sizing

- Right-size backup proxies based on workload, VM count, and data change rates
- Calculate storage capacity and performance requirements for backup repositories

 **Lab:** Size Proxy · Configure Repository · Validate Throughput

Transport Modes & Parallel Processing

- Choose the optimal transport mode (Direct SAN, HotAdd, NBD) for your environment
- Configure concurrent tasks and per-job parallelism to reduce backup windows

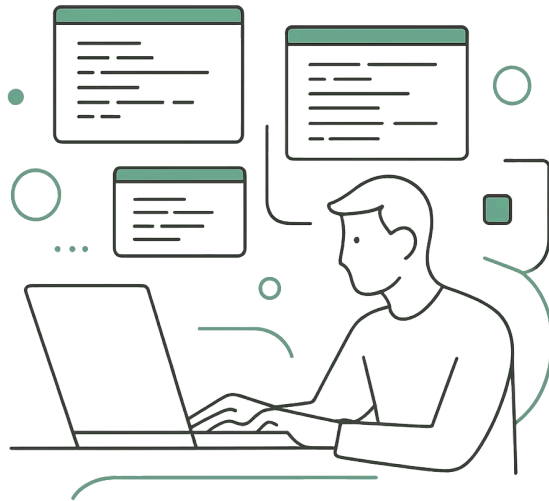
 **Lab:** Switch Transport Mode · Enable Parallel Processing · Measure Job Duration

Bottleneck Analysis & Network Optimization

- Identify and resolve CPU, network, disk, and proxy bottlenecks using Veeam ONE
- Tune WAN acceleration, compression, and deduplication for efficient data transfer

 **Lab:** Run Bottleneck Report · Enable WAN Acceleration · Compare Backup Speed

Troubleshooting (L2/L3 Focus)



Diagnose & Resolve at Enterprise Level

Master L2/L3 troubleshooting techniques for Veeam environments — from job failures to infrastructure-level issues — using logs, tools, and systematic analysis.

- **Backup Job Failures** — Analyze job logs, retry settings, and infrastructure connectivity to resolve failures
- **Snapshot & CBT Issues** — Fix VMware snapshot consolidation errors and reset Changed Block Tracking
- **Repository & Proxy Failures** — Resolve offline repositories, proxy errors, and transport mode fallbacks
- **SQL & Credential Issues** — Troubleshoot SQL connectivity, authentication failures, and permission errors
- **Network & Space Issues** — Address bottlenecks, latency problems, and storage capacity exhaustion



Lab: Simulate Backup Failure · Resolve Snapshot Error · Analyze Logs · Troubleshoot Repository Issues

Enterprise Design & Best Practices

Backup Architecture Design

Design scalable, resilient, and enterprise-grade backup solutions

Multi-Site Backup

Implement backup strategies across multiple locations to ensure data availability

DR Strategy

Develop disaster recovery plans that minimize downtime

Capacity Planning

Estimate and optimize storage, compute, and network resources for future backup workloads.

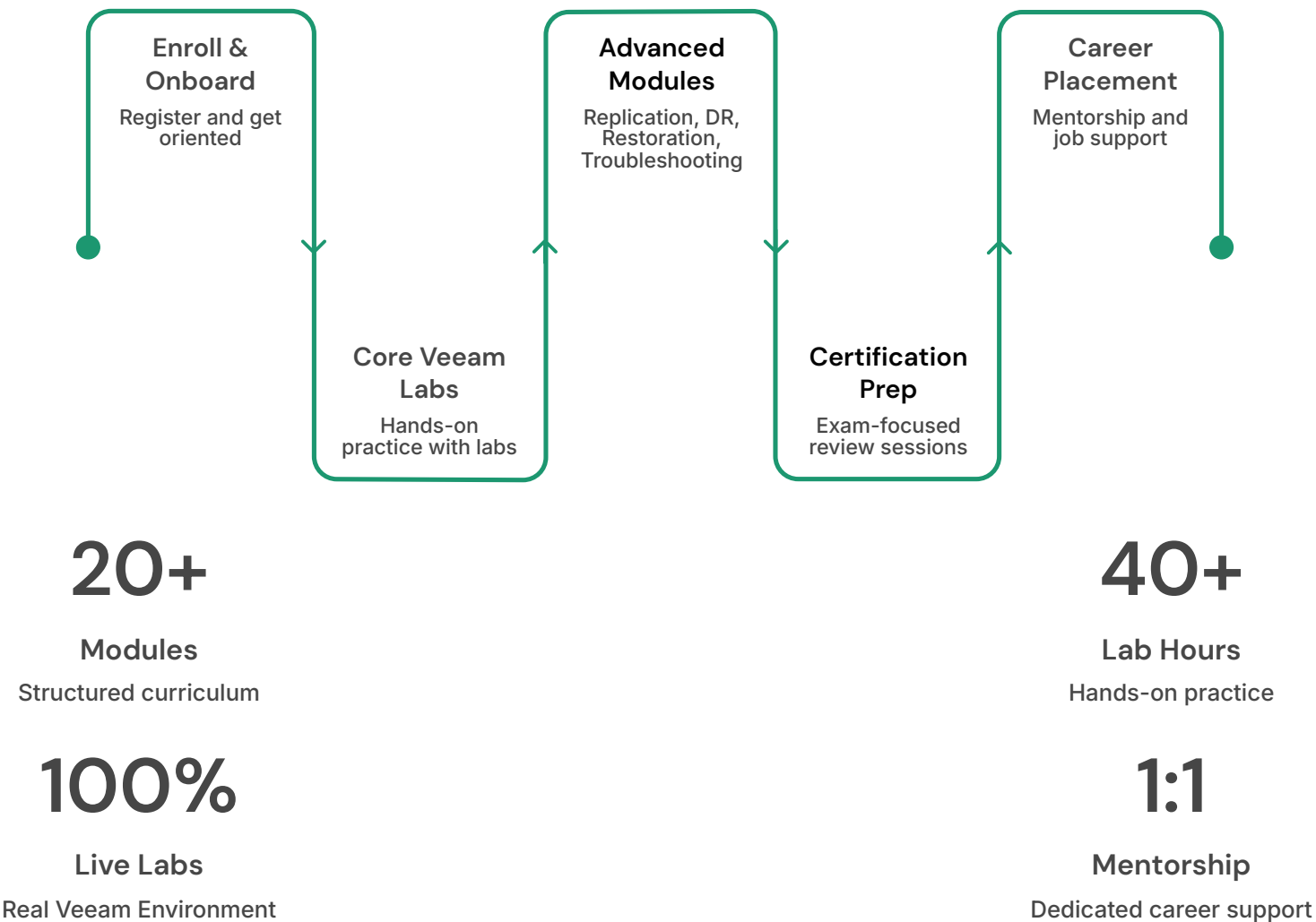
Security Hardening

Protect backup infrastructure through encryption, immutability, access controls, MFA, and security best practices.

Compliance

Meet regulatory and organizational requirements through compliant backup and data retention policies.

Your Learning Journey



Ready to Get Started?

Join SMART MATE VENTURES and build the enterprise VMware skills that top organizations demand.



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